

Kazi Md. Tawsif Rahman

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Academic Profile

Computer Science graduate and backend systems engineer conducting applied empirical research at the intersection of network security, privacy-preserving machine learning, and trustworthy evaluation. Current work develops a reproducible privacy–utility audit for machine-learning-based network intrusion detection, combining IDS-specific evaluation, shadow-model membership-inference auditing, and formally accounted DP-SGD. Published work includes machine-learning-based insurance fraud detection and an IEEE conference paper on an insurance awareness and decision-support application.

Research Interests

Primary: network security; privacy-preserving machine learning; membership-inference auditing; trustworthy evaluation of security-critical machine learning.

Supporting: dependable backend and data-intensive systems; reproducible experimental software; reliable and auditable services.

Research Direction for PhD Study

I am interested in practical security and machine-learning systems in which model utility, privacy leakage, and formal protection are evaluated under one explicit, reproducible protocol. My current study asks whether formally accounted DP-SGD can reduce measurable training-membership leakage in a tabular intrusion detector while retaining operationally acceptable Recall and false-negative rate. Long term, I want to develop deployable trustworthy learning systems whose privacy and reliability claims remain aligned with empirical evidence.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh

Sep 2024

B.Sc. in Computer Science and Engineering

CGPA: 3.59 / 4.00

Publications

1. **Journal Article** Kazi Md. Tawsif Rahman and Chowdhury Mahfuzul Hoq. “An Automated System for Detecting Property Insurance Fraud Using Machine Learning.” *International Journal of Mathematical Sciences and Computing (IJMSC)*, 2024. DOI: 10.5815/ijmsc.2024.03.02.
2. **IEEE Conference Paper** Mahfuzulhoq Chowdhury, Kazi Md. Tawsif Rahman, and Hossain Ahmad Maruf. “Whistle Blower: An Insurance Awareness Mobile Application with Insurance Policy Selection, Fraud Detection, Critical Help, Complaint Features.” *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, pp. 1–6, 2024. DOI: 10.1109/CSITSS64042.2024.10817002.

Research Experience

A Reproducible Privacy–Utility Audit for ML-Based Network Intrusion Detection

2026 –

Present

- Designed a locked NSL-KDD binary-classification protocol with validation-only threshold selection and an untouched KDDTest+ evaluation split.
- Established Random Forest, XGBoost, and MLP utility baselines; the selected non-private MLP achieved 76.91% Recall, 23.09% FNR, and 84.08% F1 on KDDTest+ at a validation-selected threshold of 0.28.
- Completed a five-shadow-model membership-inference audit with score-only and label-aware features. The strongest overall baseline attack was near chance (ROC-AUC approximately 0.5029), establishing an important floor-effect limitation for later privacy comparisons.
- Completed PyTorch parity and an Opacus DP-SGD feasibility run with explicit accounting at $\varepsilon = 7.9986$ and $\delta = 1.134 \times 10^{-5}$. The single smoke test achieved 66.20% Recall, 33.80% FNR, 76.80% F1, and 91.93% PR-AUC.
- Current work: execute the full privacy-budget sweep, apply the same membership audit to every DP model, and estimate uncertainty across repeated runs. The feasibility result is not yet evidence of reduced membership leakage.
- Repository: github.com/tawsif113/privacy-utility-dp-ids.

Insurance Fraud Detection Using Machine Learning

Published, 2024

- Developed an automated property-insurance fraud-detection approach using machine-learning methods.
- Published in *International Journal of Mathematical Sciences and Computing*; demonstrates applied ML, problem formulation, technical writing, and evaluation-oriented research.

Whistle Blower: Insurance Awareness and Decision-Support Application

Published, 2024

- Co-authored an IEEE CSITSS paper on a mobile application combining insurance awareness, policy selection, fraud detection, critical help, and complaint features.
- Research relevance: applied software systems, fraud detection, consumer-facing decision support, and human-centered financial technology.

Professional Experience

Software Engineer

Dec 2024 – Present

BRAC IT Services Ltd., Dhaka, Bangladesh

CRM Platform

- Designed and implemented backend services for accounts, contacts, leads, opportunities, quotes, campaigns, territories, products, and pipelines, delivering 168 REST endpoints across 30 controllers.
- Built configurable rule-based assignment for leads and opportunities, including priority, schedules, conditional routing, rotation rules, duplicate checks, and field/operator/value validation.
- Improved Kanban movement scalability by replacing full-column reordering with fixed-length order keys, reducing a move to a single-rank assignment.
- Implemented access-controlled Kanban operations with idempotent moves, transition validation, audit logs, and timeline entries.

Ticketing System

- Built backend services for ticket creation, search, comments, forwarding, assignment rules, teams, SLA/KPI tracking, feedback, notifications, and metadata, delivering 96 REST endpoints across 21 controllers.
- Implemented least-loaded team routing using Redis-backed workload counters and a min-heap, with absence-reliever handling for unavailable resolvers.
- Maintained resolver workload consistency across PostgreSQL and Redis through asynchronous updates after ticket creation, reassignment, and resolution.

Technical and Research Skills

Programming	Python, Java, SQL, C, C++
ML / Privacy	scikit-learn, XGBoost, PyTorch, Opacus, DP-SGD, membership-inference auditing, threshold selection
Research Methods	Literature review, experimental design, baseline reproduction, threat-model specification, privacy accounting, reproducibility manifests, result analysis, technical writing
Backend / Systems	Spring Boot, Spring Security, Spring Data JPA, Hibernate, Spring Modulith, REST APIs, Gradle, Docker, Linux
Data / Messaging	PostgreSQL, MongoDB, Redis, RabbitMQ, Flyway, JSONB, asynchronous processing, retry handling
Testing / Quality	JUnit 5, Mockito, API testing, Git, reproducible development workflows

Awards and Achievements

- Codeforces Specialist; reached a rating above 1500.
- Finalist, CUET IUPC 2024.
- Finalist, CUET IUPC 2023.
- Finalist, CUET Tech Carnival 2020.

Selected Coursework

Algorithms and Data Structures; Database Systems; Computer Networks; Operating Systems; Software Engineering; Artificial Intelligence; Machine Learning; Information Security.